Dhana Karthikeya Ventrapragada

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Education

University of California, Berkeley **Expected Dec 2027**

Bachelor of Arts in Data Science and Applied Mathematics | GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Structure & Interpretation of Computer Programs, Data Structures; *In progress:* Functional Programming & AI-Assisted App Development, Data Engineering

Technical Skills

Languages: Python, Java, SQL, TypeScript, Gleam, LaTeX

Engineering: FastAPI, React, REST APIs, PostgreSQL (query plans, indexing), MongoDB, Auth0, AWS S3, Supabase, Apify, Git

Fundamentals: Data Structures, Graph Algorithms, Complexity Analysis, OOP, Functional Programming

ML & Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, Claude API, Factory Droid SDK

Experience

Guang Labs | Engineering Intern | Remote **May 2026 - Aug 2026**

- Engineered a backend REST API service for an AI video startup that classifies a user's image or video by content type (sports, gaming, talking) using the Gemini API.
- Architected an asynchronous submit-and-poll workflow that processes long videos in the background rather than blocking users (FastAPI, PostgreSQL, Auth0).
- Delivered the service end to end after code review, supporting uploads from **4** sources (local, S3, web links) and validating it with **86** automated tests.
- Developed an in-app notification bell, from the database to the React interface, that alerts users when their video finishes processing or fails, and ensured failures never disrupt processing or send duplicate alerts on retries.

IDX Exchange | Data Science Intern | Remote **Jun 2026 - Aug 2026**

- Engineered a pipeline that predicts California home sale prices from **143K** real-estate records, typically within **8.6%** of the actual price.
- Benchmarked four model types and selected a tuned XGBoost (R^2 of **0.88**), resolving a data leakage bug and cutting a simpler model's error from **29%** to **15%** with a school district feature.
- Analyzed errors by price band to find weak spots: luxury homes (**13%** error vs. **6.5%** mid-priced) and low-sales towns.
- Retrained the final model from scratch and matched results exactly to confirm reproducibility.

Projects

Hindsight (Cal Hacks 13.0) | *Claude, FastAPI, React, Supabase, Apify, Factory Droid* GitHub

- Co-designed a transfer planner: a **FastAPI** backend that matches courses to official ASSIST agreements with deterministic rules (minimum grades, agreement years), a **React** and **TypeScript** interface, and **Supabase**.
- **Claude** reads transcripts and class schedules that students confirm and answers with numbered citations that code validates; **Apify** scrapes campus pages and a **Factory Droid** agent repairs only broken scraper scripts, in a sandbox.
- Tested with **10** transfer students, who said it would have helped during their own transfer.

Resume Tailor | *Python, FastAPI, LaTeX, Claude* GitHub

- Built with **Claude**: a **Python** and **FastAPI** app that tailors a **LaTeX** resume to a job posting.
- **Claude API** rewrites must quote my own records; code rejects invented numbers and checks page fit and ATS parsing.

Build Your Own World (Dungeon Game) | *Java* GitHub

- Built a seeded, procedurally generated dungeon game in Java where enemies chase the player along shortest paths (BFS), moving two steps for every one the player takes once in range.
- Designed a quiz-battle system with **100+** CS 61B trivia questions: enemy contact drains health until the player answers correctly, and health packs restore it.

Leadership & Involvement

UC Berkeley CDSS | Undergraduate Course Staff, CDSS 198 | Berkeley, CA **Aug 2026 - Present**

- Run exam reviews and write practice worksheets for CS and data science classes for transfer students, and give resume feedback.

UC Berkeley | Undergraduate Course Staff, CS 61A | Berkeley, CA **Jun 2026 - Aug 2026**

- Taught **4** weekly exam-prep sections in Python, Scheme, and SQL, reinforcing data structures like linked lists, trees, lists, and dictionaries, held office hours, and helped write midterm and final questions.